MAST20030 Differential Equations
Supplementary Material

Trigonometric and hyperbolic formulae

$$\sin(x) = \frac{1}{2\,\mathrm{i}}\left\{\mathrm{e}^{\mathrm{i}\,x} - \mathrm{e}^{-\mathrm{i}\,x}\right\}, \qquad \sinh(x) = \frac{1}{2}\left\{\mathrm{e}^{x} - \mathrm{e}^{-x}\right\}$$

$$\cos(x) = \frac{1}{2}\left\{\mathrm{e}^{\mathrm{i}\,x} + \mathrm{e}^{-\mathrm{i}\,x}\right\}, \qquad \cosh(x) = \frac{1}{2}\left\{\mathrm{e}^{x} + \mathrm{e}^{-x}\right\}$$

$$\cos^2(x) + \sin^2(x) = 1, \qquad \cosh^2(x) - \sinh^2(x) = 1$$

$$1 + \tan^2(x) = \sec^2(x), \qquad 1 - \tanh^2(x) = \mathrm{sech}^2(x)$$

$$\cot^2(x) + 1 = \mathrm{cosec}^2(x), \qquad \coth^2(x) - 1 = \mathrm{cosech}^2(x)$$

$$\sin(x+y) = \sin(x)\,\cos(y) + \cos(x)\,\sin(y), \qquad \sinh(x+y) = \sinh(x)\,\cosh(y) + \cosh(x)\,\sinh(y)$$

$$\cos(x+y) = \cos(x)\,\cos(y) - \sin(x)\,\sin(y), \qquad \cosh(x+y) = \cosh(x)\,\cosh(y) + \sinh(x)\,\sinh(y)$$

$$\sin(x)\,\sin(y) = \frac{1}{2}\left\{\cos(x-y) - \cos(x+y)\right\}, \qquad \sinh(x)\,\sinh(y) = \frac{1}{2}\left\{\cosh(x+y) - \cosh(x-y)\right\}$$

$$\cos(x)\,\cos(y) = \frac{1}{2}\left\{\cos(x-y) + \cos(x+y)\right\}, \qquad \cosh(x)\,\cosh(y) = \frac{1}{2}\left\{\cosh(x+y) + \cosh(x-y)\right\}$$

$$\sin(x)\,\cos(y) = \frac{1}{2}\left\{\sin(x-y) + \sin(x+y)\right\}, \qquad \sinh(x)\,\cosh(y) = \frac{1}{2}\left\{\sinh(x+y) + \sinh(x-y)\right\}$$

Limits

$$\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad (p > 0), \qquad \lim_{n\to\infty} r^n = 0 \quad (|r| < 1)$$

$$\lim_{n\to\infty} a^{1/n} = 1 \quad (a > 0), \qquad \lim_{n\to\infty} n^{1/n} = 1$$

$$\lim_{n\to\infty} \frac{a^n}{n!} = 0 \quad (\text{all } a), \qquad \lim_{n\to\infty} \frac{\ln(n)}{n^p} = 0 \quad (p > 0)$$

$$\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = \mathrm{e}^a \quad (\text{all } a), \qquad \lim_{n\to\infty} \frac{n^p}{a^n} = 0 \quad (\text{all } a, p > 1)$$

Series

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \text{is} \quad \begin{cases} \text{convergent if } p > 1 \\ \text{divergent if } p \le 1 \end{cases}$$

$$\sum_{n=1}^{\infty} a\,r^n \quad \text{is} \quad \begin{cases} \text{convergent if } |r| < 1 \\ \text{divergent if } |r| \ge 1 \end{cases}$$

Reduction of order formula for second order ODEs

$$y''(t) + f_1\, y'(t) + f_0\, y(t) = g(t) \quad \Rightarrow \quad y = v \int \left[\frac{1}{I} \int \frac{Ig}{v}\, \mathrm{d}t \right] \mathrm{d}t \quad \text{where} \quad I = v^2 \mathrm{e}^{\int f_1\, \mathrm{d}t}$$

and $y(t) = u(t)\, v(t)$ for given complementary solution $v(t)$.

Laplace transform formulae

$$F(s) = \mathcal{L}\{f(t)\} = \int_0^\infty f(t)\, \mathrm{e}^{-s\,t}\, \mathrm{d}t, \qquad f(t) = \mathcal{L}^{-1}\{F(s)\}$$

$$\mathcal{L}\{e^{at} f(t)\} = F(s-a), \qquad \mathcal{L}\{f(t-a)H(t-a)\} = e^{-as}F(s)$$

$$\mathcal{L}\{af(t) + bg(t)\} = a\mathcal{L}\{f(t)\} + b\mathcal{L}\{g(t)\}$$

$$\mathcal{L}\{f^{(n)}\} = s^n\mathcal{L}\{f\} - s^{n-1}f(0) - \ldots - f^{(n-1)}(0)$$

$$\mathcal{L}\{f * g\} = \mathcal{L}\{f\}\mathcal{L}\{g\} \quad \text{where} \quad (f*g)(t) = \int_0^t f(t-\tau)\, g(\tau)\, \mathrm{d}\tau$$

Laplace Transform table

$F(s) = \mathcal{L}\{f(t)\}$	$f(t)$
$1/s$	1
$1/s^n \quad (n = 1, 2, \ldots)$	$t^{n-1}/(n-1)!$
$1/(s-a)$	e^{at}
$\omega/(s^2+\omega^2)$	$\sin \omega t$
$s/(s^2+\omega^2)$	$\cos \omega t$
$a/(s^2-a^2)$	$\sinh at$
$s/(s^2-a^2)$	$\cosh at$
e^{-as}	$\delta(t-a)$

Fourier series formulae

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} \{a_n \cos(n\,\pi\,x\,/\,L) + b_n \sin(n\,\pi\,x\,/\,L)\}$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\cos(n\,\pi\,x\,/\,L)\;\mathrm{d}x \quad (n = 0, 1, 2, \ldots)$$

$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\,\sin(n\,\pi\,x\,/\,L)\;\mathrm{d}x \quad (n = 1, 2, \ldots)$$

Fourier transform formulae

$$\widehat{f}(k) = \mathcal{F}\{f\} = \int_{-\infty}^{\infty} f(x)\,\mathrm{e}^{-\mathrm{i}\,k\,x}\;\mathrm{d}x, \qquad f(x) = \mathcal{F}^{-1}\{\widehat{f}\} = \frac{1}{2\,\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\,\mathrm{e}^{\mathrm{i}\,k\,x}\;\mathrm{d}k$$

$$\mathcal{F}\left\{\frac{d^n f}{dx^n}\right\} = (\mathrm{i}\,k)^n\,\widehat{f}, \qquad \mathcal{F}\left\{x^n\,f\right\} = \mathrm{i}^n\,\frac{d^n\,\widehat{f}}{dk^n}$$

$$\mathcal{F}\{f * g\} = \mathcal{F}\{f\}\mathcal{F}\{g\} \quad \text{where} \quad (f * g)(x) = \int_{-\infty}^{\infty} f(x-\xi)\,g(\xi)\;\mathrm{d}\xi$$